

Joint-Liner Interaction Tutorial

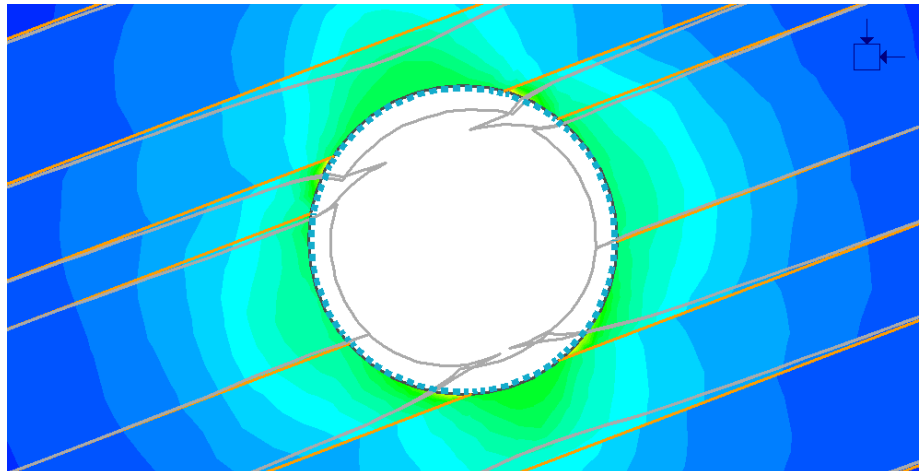
This tutorial demonstrates how to model liner support in a jointed rock mass, when joints intersect excavation boundaries on which liner support will be installed. In order to correctly model the interaction of the joint-liner intersections, we must define a Composite Liner which includes a joint at the liner-rock interface.

In this case, the liner will resist slip on the joints such that it remains intact and continuous around the excavation.

The analysis will be conducted in two parts. The first part shows the response of a tunnel in jointed rock without a liner. The second part shows the effect of adding the liner support.

Topics Covered

- Rock joints
- Composite Liner with joint
- Generalized Hoek-Brown failure criterion
- Copying boundaries and relative coordinates
- Graph joint data
- Liner bending moments



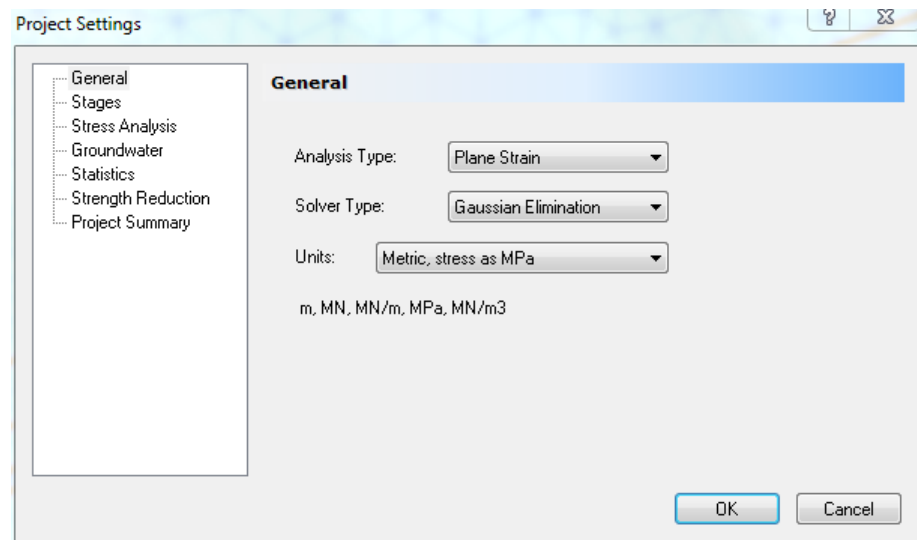
Model

Start the *Phase2* Model program.

Project Settings



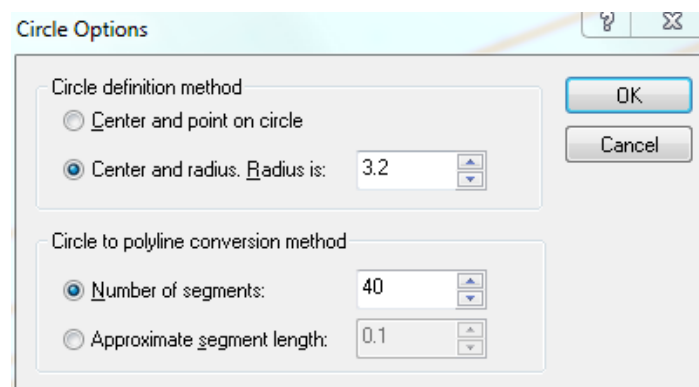
Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Set the Number of Stages to 2. In this analysis, the first stage will bring the unexcavated rock to equilibrium and the tunnel will be excavated in the second stage. Define the units as being “Metric, stress as MPa”. Click OK to close the Project Settings dialog. If you see a warning about the unit system then hit OK.



Boundaries



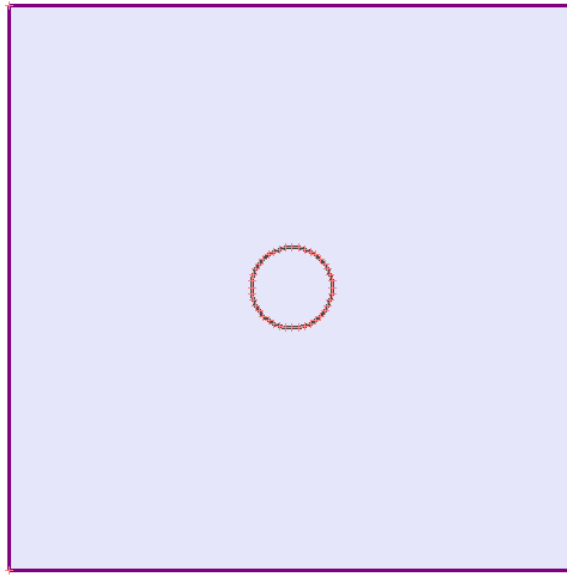
First we will define the excavation. Select **Add Excavation** from the **Boundaries** menu. Type the letter **i** to indicate you wish to draw a circle (or right click and choose Circle) and hit Enter. You will now see the dialog for entering a circle. Select the **Centre and radius** option and set the radius to 3.2. Set the Number of segments to 40 as shown.



Click OK. You will now see a circle that you can drag around with the mouse. Enter 0,0 for the centre coordinates and hit Enter. The excavation geometry is now defined.



To define the external boundary, select **Add External** from the **Boundaries** menu. The default boundary is a box around the excavation and the default expansion factor is 3. Click OK to accept these defaults. The model should now look like this.



Joints



From the **Boundaries** menu, select **Add Joint**. You will see the Add Joint dialog, which allows you to select a Joint property type, end condition and installation stage. We will use the default selections, so just select OK.

NOTE: see the *Phase2* Help system for a discussion of the Joint End Condition option.

Now enter the following coordinates defining the joint.

-23 , -17

23 , 0.5

Enter

The joint is now added to the model. Note that the “closed” Joint End Condition is indicated by an icon of a circle with a triangle inside, at both ends of the joint.

Note that the two points defining the joint were actually entered just outside of the external boundary, and *Phase2* automatically intersected the boundaries and added new vertices.



We now wish to generate a series of parallel joints. The easiest way to accomplish this is to “copy and paste” the original joint. To do this, right click on the joint and select **Copy Boundary** from the resulting menu. You can now enter relative coordinates to make a copy of the joint shifted by some amount. At the prompt enter:

@ 0 , 3

This will create a copy of the joint shifted 0 m in the x direction and 3 m in the y direction.

Repeat these steps by right clicking on the *original* joint each time and entering the following relative coordinates:

@ 0 , 5

@ 0 , 7

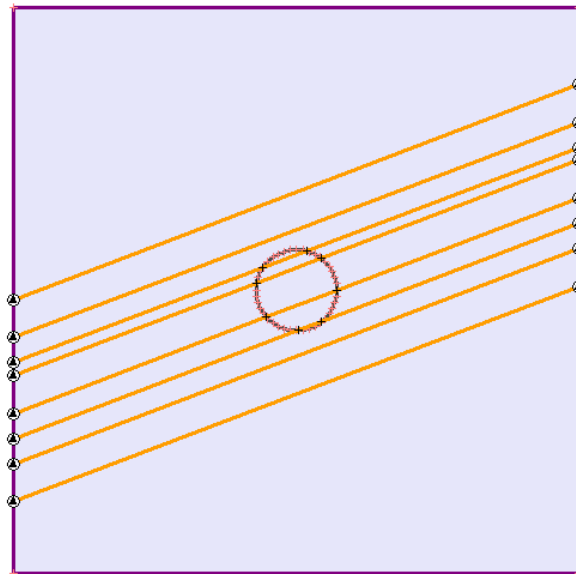
@ 0 , 10

@ 0 , 11

@ 0 , 13

@ 0 , 16

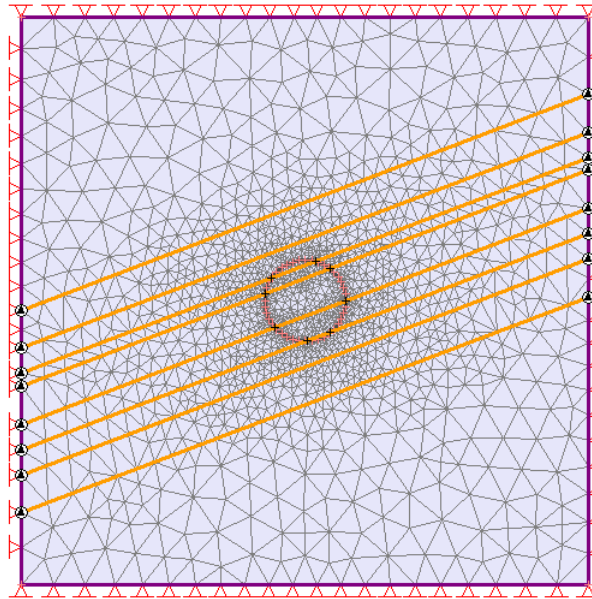
Your model should now appear as shown.



Mesh



Now that all of the boundaries have been defined we can generate the finite element mesh. Select the **Mesh Setup** option in the **Mesh** menu. The default options should be sufficient for this model. Ensure that the Mesh Type is Graded, the Element Type is 3 Noded Triangles, the Gradation Factor is 0.1 and the Default Number of Nodes on All Excavations is 75. Click the Discretize button and then the Mesh button. Click OK to close the dialog. The model should now look like this:



Field Stress



Select **Field Stress** from the **Loading** menu. The tunnel is assumed to be deep underground and the stress is assumed to be caused by the overburden. Select Gravity for the Field Stress Type and enter 1100 m for the Elevation. All other values can be left as default values. The dialog should appear as shown.

Field Stress Properties

Field Stress Type: Gravity

☐ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (m): 1100

Unit Weight of Overburden (MN/m³): 0.027

Total Stress Ratio (horiz/vert in plane): 1

Total Stress Ratio (horiz/vert out-of-plane): 1

Locked-in horizontal stress (in plane) (MPa, Comp. +): 0

Locked-in horizontal stress (out-of-plane) (MPa, Comp. +): 0

OK Cancel Statistics... Advanced >>

Click OK to close the dialog.

Material Properties



Select **Define Materials** from the **Properties** menu. For Material 1, change the name to Graphitic Phyllite. For Initial Element Loading select Field Stress & Body Force. Leave the Unit Weight and Poisson's Ratio at their default values (0.027 MN/m³ and 0.3, respectively). For Young's Modulus enter 1645 MPa. Under Strength Parameters select Generalized Hoek-Brown for the Failure Criterion. Set the Material Type to be Plastic. Enter the Generalized Hoek-Brown parameters as shown below:

The screenshot shows the 'Define Material Properties' dialog box for a material named 'Graphitic Phyllite'. The dialog is organized into several sections:

- Material List:** A list of materials from Material 2 to Material 29 is shown on the left. 'Graphitic Phyllite' is selected at the top.
- General Properties:**
 - Name: Graphitic Phyllite
 - Material Color: (blue swatch)
 - Initial Element Loading: Field Stress & Body Force
 - Unit Weight: (MN/m³): 0.027
- Elastic Properties:**
 - Elastic Type: Isotropic
 - Poisson's Ratio: 0.3
 - Young's Modulus (MPa): 1645
 - Young's Modulus (resid) (MPa): 20000
 - E1 (MPa): 20000, E2 (MPa): 20000, E3 (MPa): 20000
 - v12: 0.2, v13: 0.2, v23: 0.2
- Strength Parameters:**
 - Failure Criterion: Generalized Hoek-Brown
 - Material Type: Plastic
 - Intact Comp. Strength (MPa): 25
 - Dilation Parameter: 0
 - mb Parameter (peak): 0.821, mb Parameter (resid): 0.821
 - s Parameter (peak): 0.0004, s Parameter (resid): 0.0004
 - a Parameter (peak): 0.522, a Parameter (resid): 0.522
- Advanced Options:**
 - Stage Properties: (checkbox) Define Factors...
 - Datum Dependent: (checkbox) Define Properties...
 - Unsaturated Shear Strength:
 - Phi b: 0
 - Air Entry (MPa): 0
- Buttons:** Copy To..., Statistics..., Show only properties used in model, OK, Cancel.

Click OK to close the dialog.

NOTE: see the *Phase2* Help system for a discussion of the meaning of the different Generalized Hoek-Brown parameters.

Joint Properties



Select **Define Joints** from the **Properties** menu. Change the name of Joint 1 to Rock Joints. For Criterion, select Mohr-Coulomb and change the friction angle to 20 degrees. Leave all other values as default. Note that leaving the Initial Joint Deformation option turned on means that the joints will deform due to the field stress as well as stresses induced by excavations. The dialog should appear as below:

The screenshot shows the 'Define Joint Properties' dialog box for 'Rock Joints'. The left pane lists various joint types, with 'Rock Joints' selected. The main area contains the following settings:

- Name:** Rock Joints
- Color:** Orange
- Slip Criterion:** Mohr-Coulomb
- Stiffness:** Normal Stiffness (MPa/m): 100000, Shear Stiffness (MPa/m): 10000
- Tensile Strength (+MPa):** 0
- Peak Cohesion (+MPa):** 0
- Peak Friction Angle (deg):** 20
- Residual Strength:** (unchecked)
 - Res. Tensile Strength (+MPa): 0
 - Res. Cohesion (+MPa): 0
 - Res. Friction Angle (deg): 20
- Initial Joint Deformation:** (checked)
- Additional Pressure Inside Joint:**
 - Groundwater Pore Pressure: (unchecked)
 - Additional Pressure (+MPa): 0
 - Apply Pressure to Liner Side Only: (unchecked)
- Stage Joint Properties:**
 - Define Factors... (button)
 - Allow slip starting from stage: 1

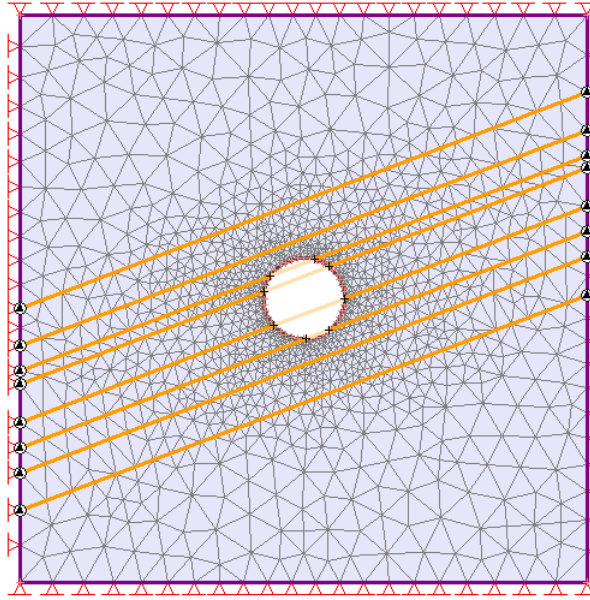
Buttons at the bottom include 'Copy To...', 'Statistics...', 'Show only properties used in model' (checkbox), 'OK', and 'Cancel'.

Click OK to close the dialog.

Excavation



The tunnel is to be excavated in the second stage so click on the Stage 2 tab at the bottom of the screen. From the **Properties** menu select **Assign Properties**. From the Assign Properties dialog, select **Excavate**. Because of the joint boundaries passing through the tunnel, the easiest way to excavate all sections of the tunnel is by using a selection window. Click and hold down the left mouse button at a point above and to the left of the tunnel. Drag the mouse to draw a box completely enclosing the tunnel. Release the button and all sections of the tunnel should be excavated as shown.



You can also excavate the tunnel by clicking inside each section that is separated by joints – but you must be careful to click inside all sections.

You have now completed modelling the tunnel without support. Save the model by choosing Save As from the File menu.

Compute

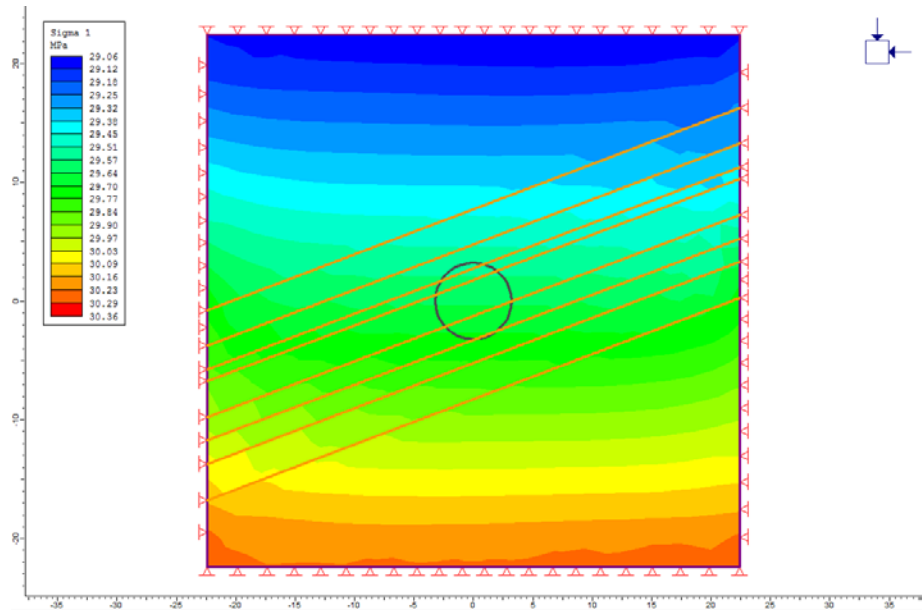


Run the model by pressing the Compute button on the toolbar. The analysis may take several minutes to run since the unsupported tunnel will experience extensive failure and large deformations.

Once the model has finished computing (Compute dialog closes), click the Interpret button to view the results.

Interpret (unsupported)

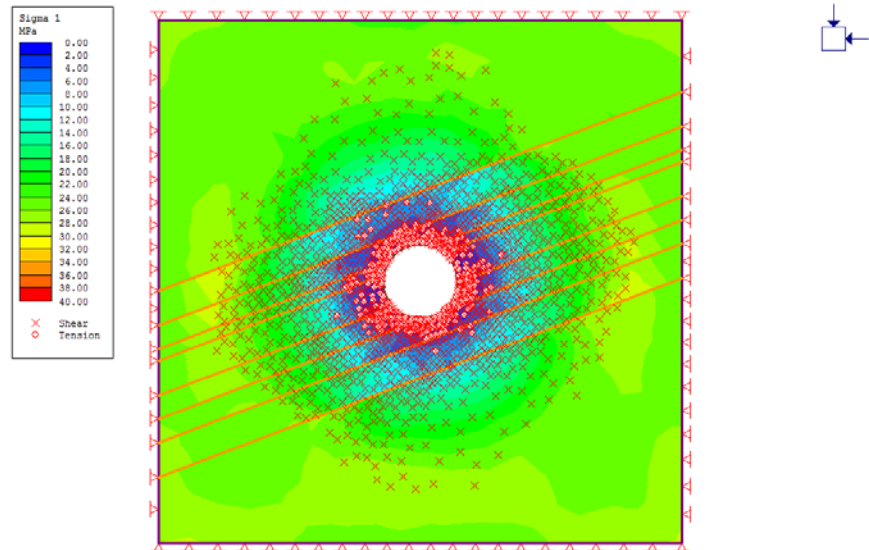
After you select the Interpret option, the *Phase2* Interpret program starts and reads the results of the analysis. You should see a screen similar to the following that shows the maximum compressive stress for Stage 1.



You can see that stress generally increases with depth as expected. There are some discontinuities in stress across the joints, however the variations in observed stress are small.



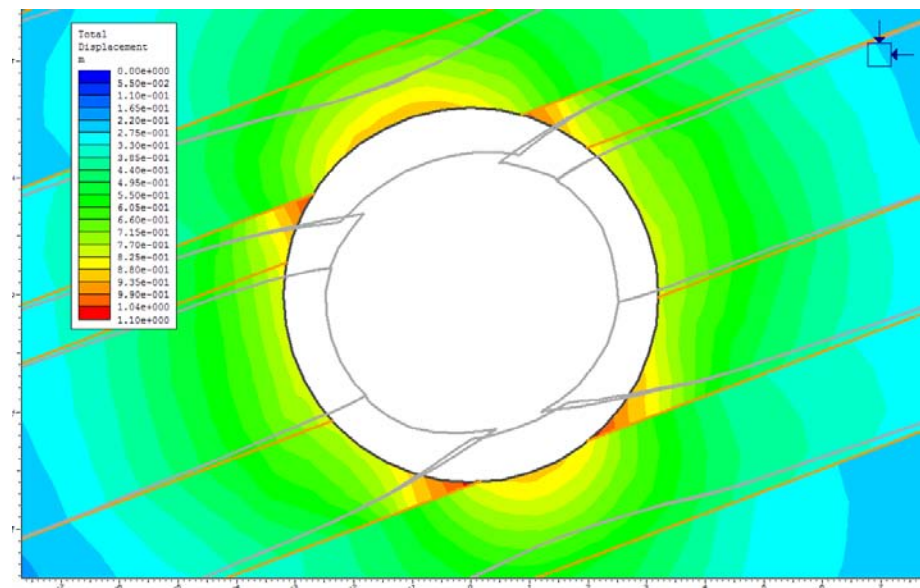
Click on the Stage 2 tab. You will now see low stresses around the tunnel with higher stresses further out. This suggests that the rock around the tunnel has failed and cannot support high stresses. Confirm this by plotting the failed elements using the **Display Yielded Elements** button on the toolbar. You can also plot the sections of joints that have yielded by clicking on the **Display Yielded Joints** button. The model should appear as follows.



There is obviously extensive failure around the tunnel that extends a significant distance into the rock mass. Also, most of the joints close to the tunnel have failed.



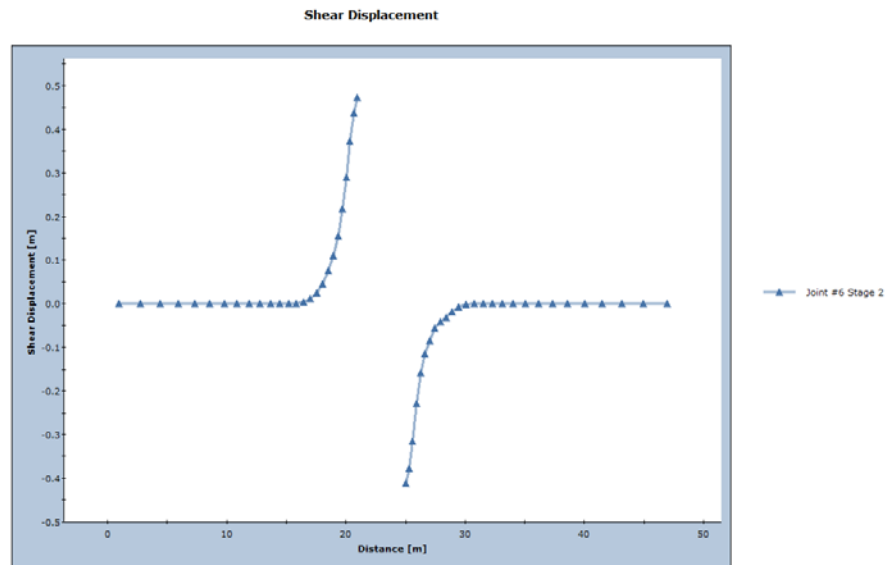
Now plot the deformation by changing the contours to Total Displacement and clicking on the **Display Deformed Boundaries** button. Turn off the Yielded Elements and zoom in on the excavation. The model will appear as shown.



You can see how the tunnel has been squeezed under stress and also how its shape has changed to become more elliptical. The joints are also showing some slip, as can be observed from the offset between opposite sides of each joint, where each joint intersects the tunnel boundary.



You can examine the slip on the joints by plotting the shear displacement. Right click on the joint that intersects the top of the tunnel. Select **Graph Joint Data**. For the Vertical Axis select Shear Displacement and click Create Plot. The graph should look like this:



This plot shows almost 10 cm of slip on the joint near the tunnel surface. NOTE: the “gap” in the joint displacement graph (no data points) is due to the excavated section of the joint passing through the tunnel.

We now wish to minimize the deformation and failure in the tunnel by adding support in the form of a shotcrete liner.

Liner Support

Go back to the *Phase2* Model program. Open the saved file from the previous part of this tutorial if necessary. We will use the same model as before but now we will add liner support and observe the effect.

Modeling Joint-Liner Interaction

To summarize the model so far – we have an excavation which is intersected by rock joints, and the excavation requires liner support in order to prevent collapse.

IMPORTANT!!! In order to correctly model the interaction of the joint-liner intersections, we must define a **Composite Liner** which includes a **joint** at the liner-rock interface.

As you will see when you plot the liner forces, this correctly models the shear force which is applied to the liner by the differential slip of the joint endpoints at the joint-tunnel intersections.

Composite Liner Properties

For the purpose of this example, it will be sufficient to define a Composite Liner which is composed of a single liner and a joint.

First we will specify the properties of the single liner. Select **Define Liners** from the **Properties** menu. Change the Name of Liner 1 to Shotcrete and leave all other values as the default. The dialog should look like this:

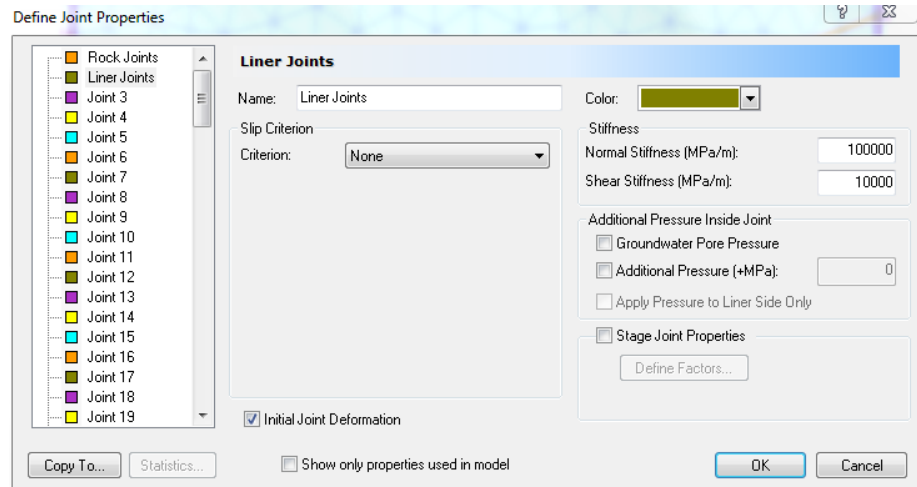


The screenshot shows the 'Define Liner Properties' dialog box for a material named 'Shotcrete'. The dialog is organized into several sections:

- Left Panel:** A list of materials including Shotcrete, Liner 2 through Liner 25. 'Shotcrete' is selected.
- Name:** Shotcrete
- Color:** A blue color swatch.
- Liner Type:** Standard Beam
- Elastic Properties:**
 - Young's Modulus (MPa): 30000
 - Poisson's Ratio: 0.2
- Strength Parameters:**
 - Material Type: Elastic (selected), Plastic
 - Compressive Strength (peak) (MPa): 35
 - Compressive Strength (residual) (MPa): 5
 - Tensile Strength (peak) (MPa): 5
 - Tensile Strength (residual) (MPa): 0
- Geometry:**
 - Thickness (m): 0.1
 - Area (m2): 0.1
 - Moment of Inertia (m4): 8.3e-005
- Include Weight in Analysis:**
 - Unit Weight (MN/m3): 0.02
- Pre-Tensioning:**
 - Pre-Tensioning Force (MN): 0
- Sliding Gap:**
 - Strain at Locking: 5 %
- Beam Element Formulation:** Timoshenko
- Buttons:** Copy To..., Define Factors..., OK, Cancel
- Checkbox:** Show only properties used in model

Click OK to close the dialog.

Now we need to define the properties of the joint between the liner and the rock. Select **Define Joints** from the **Properties** menu. Click on the tab for Joint 2. Change the name to Liner Joints and leave all other selections as default values as shown.



The 'Define Joint Properties' dialog box is shown with the 'Liner Joints' tab selected. The left pane lists various joint types, with 'Liner Joints' highlighted. The main area shows the following settings:

- Name:** Liner Joints
- Color:** (Yellow)
- Slip Criterion:** Criterion: None
- Stiffness:**
 - Normal Stiffness (MPa/m): 100000
 - Shear Stiffness (MPa/m): 10000
- Additional Pressure Inside Joint:**
 - ☐ Groundwater Pore Pressure
 - ☐ Additional Pressure (+MPa): 0
 - ☐ Apply Pressure to Liner Side Only
- ☐ Stage Joint Properties (with a 'Define Factors...' button)
- ☒ Initial Joint Deformation

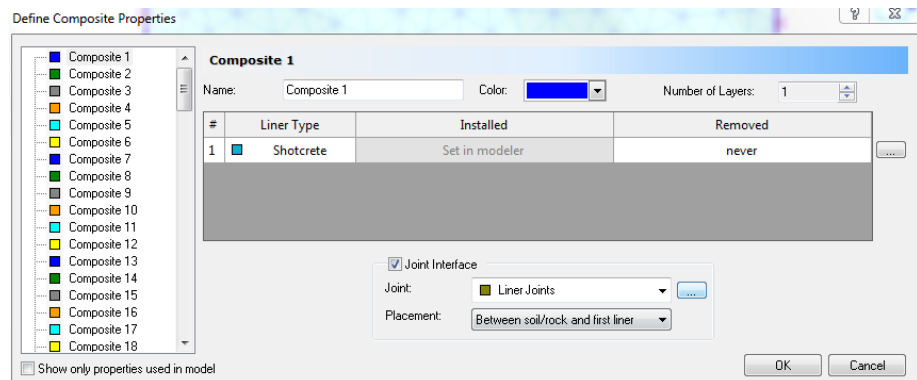
Buttons at the bottom include 'Copy To...', 'Statistics...', 'Show only properties used in model', 'OK', and 'Cancel'.

Click OK to close the dialog.



Now we can set up the composite liner. Select **Define Composite** from the **Properties** menu.

Our composite layer is to be made up of the shotcrete layer and a joint. The “Number of Layers” box should be set to 1. Set the **Liner Type** pull-down menu to **Shotcrete**. Ensure the **Joint Interface** checkbox is active and the **Joint** pull-down menu is set to **Liner Joints**. The dialog should look like this:



The 'Define Composite Properties' dialog box is shown with the 'Composite 1' tab selected. The left pane lists various composite types, with 'Composite 1' highlighted. The main area shows the following settings:

- Name:** Composite 1
- Color:** (Blue)
- Number of Layers:** 1
- Table:**

#	Liner Type	Installed	Removed
1	Shotcrete	Set in modeler	never
- ☒ **Joint Interface**
 - Joint:** Liner Joints
 - Placement:** Between soil/rock and first liner

Buttons at the bottom include 'Show only properties used in model', 'OK', and 'Cancel'.

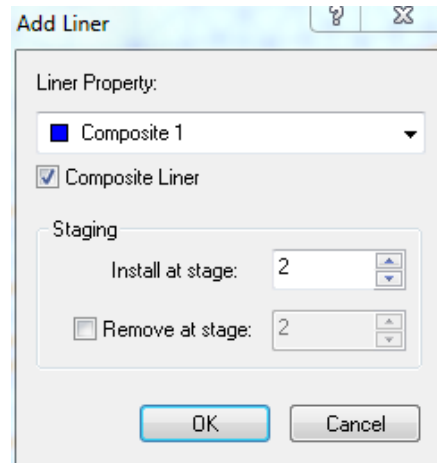
Click OK to close the dialog.

NOTE: it is very important in this model that you use a composite liner with a joint. If you only use a single liner then it will not resist slip on the rock joints and the liner will segment and become discontinuous around the tunnel.

Add Support

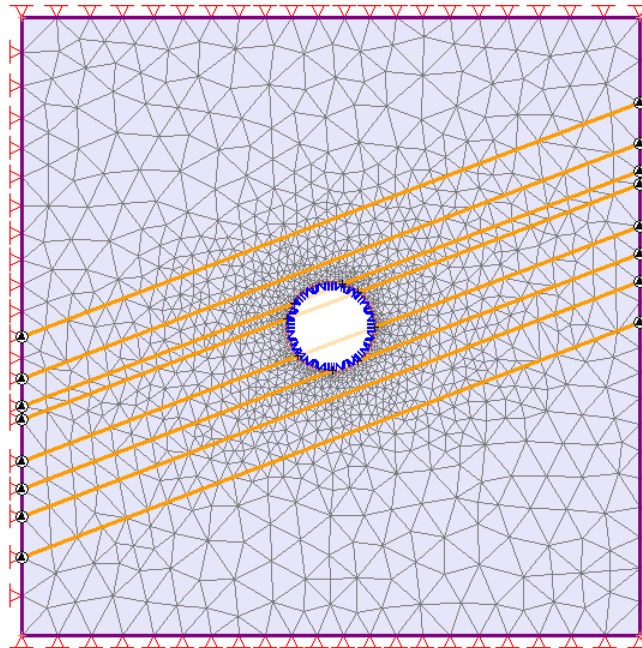


In this model, we will add the liner in Stage 2. To add the composite liner, first go to Stage 2. Select **Add Liner** from the **Support** menu. In the Add Liner dialog, make sure the **Composite Liner** checkbox is selected. The Liner Property should be Composite 1. The value for **Install at stage** should be 2 as shown.



Click OK to close the dialog. Now select all of the segments that make up the tunnel, by clicking and dragging a selection window (hold down the left mouse button and drag a window to encompass the entire tunnel). Hit Enter to finish selection.

Your model should look like this for Stage 2.



Your model is now finished. Save your model by choosing Save As from the File menu.

Compute



Run the model by pressing the **Compute** button on the toolbar. The analysis should take under a minute to run.

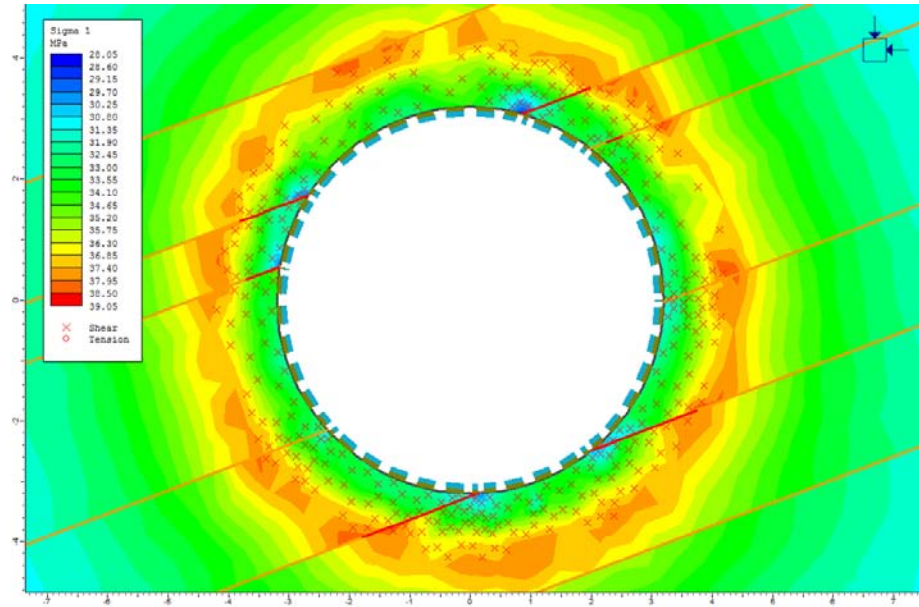


Once the model has finished computing (Compute dialog closes), click the Interpret button to view the results.

Interpret (with support)



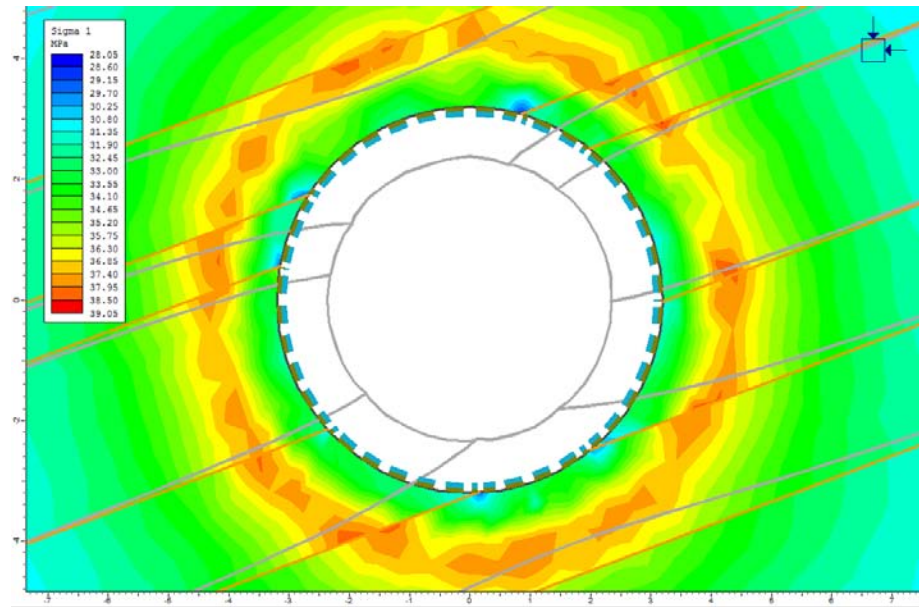
The model behaviour for Stage 1 will be the same as before. Select the Stage 2 tab. You will now see a ring of high stress around the tunnel but slightly away from the tunnel boundary. This suggests that the rock directly adjacent to the boundary has failed and cannot support high stresses. Confirm this by plotting the failed elements using the **Display Yielded Elements** button on the toolbar. You can also plot the sections of joints that have yielded by clicking on the **Display Yielded Joints** button. Zoom in on the excavation and it should appear as shown.



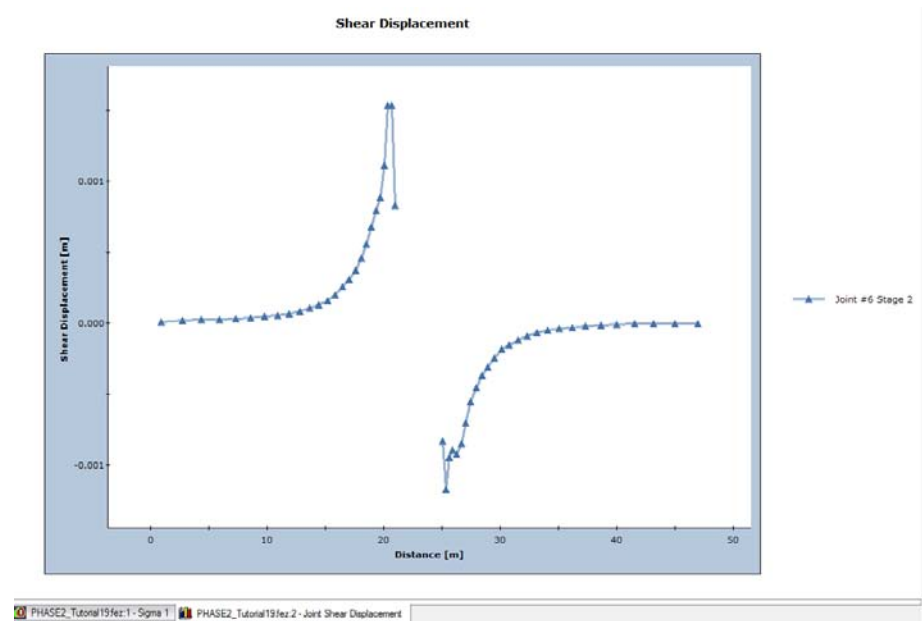
You can see that elements around the tunnel have failed in shear and also that joints near the top and bottom of the tunnel have failed (shown as red lines). However the failure of elements and joints is much less severe than observed in the unsupported model. Note that none of the liner elements can fail since we set the liner material type to be elastic.



Now plot the deformation by changing the contours to Total Displacement and clicking on the **Display Deformed Boundaries** button. Turn off the Yielded Elements and Yielded Joints. You can see that the tunnel (and liner) have displaced inwards and that there is little slip on the joints compared with the amount of tunnel closure. The tunnel has also maintained its circular shape.

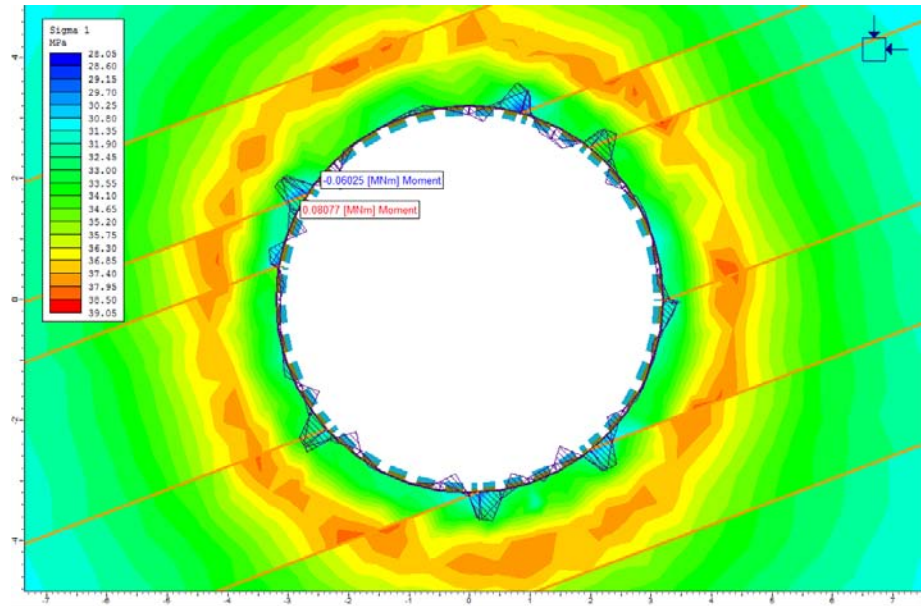


The amount of slip on the joints is small but it is not zero. You can examine the slip on the joints by plotting the displacement. Right click on the joint that intersects the top of the tunnel. Select **Graph Joint Data**. For the Vertical Axis select Shear Displacement and click Create Plot. The graph should look like this:



It is clear that the amount of slip is increasing as the joint approaches the tunnel, however, the slip on the joint is about 50 times less than the slip observed in the unsupported tunnel.

Now examine the behaviour of the liner. Go back to the window showing the tunnel in Stage 2. Turn off the deformed boundaries display. Right click on the liner and select Show Values → Bending Moment. The screen should look like this:

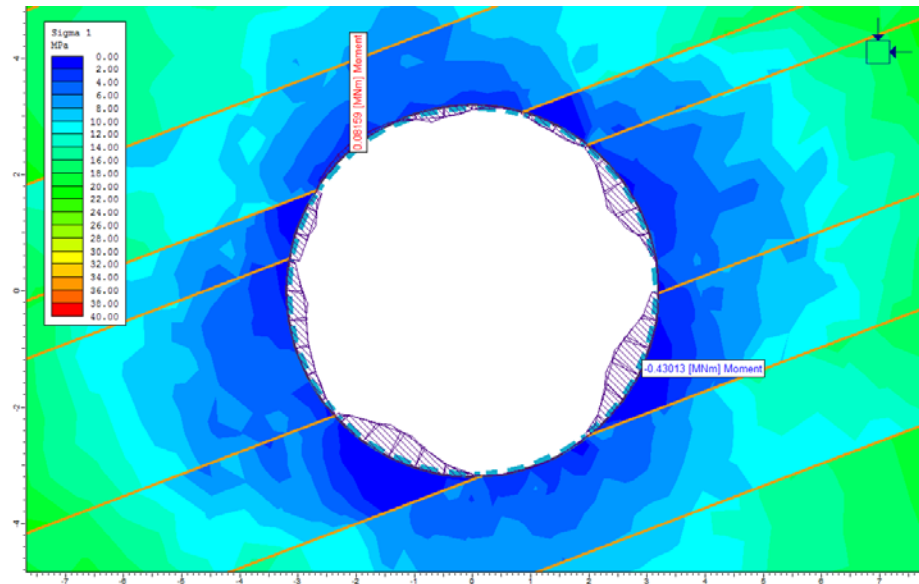


You can see that there are very large bending moments where the liner is intersected by the rock joints. The joints are trying to slip but they are being resisted by the liner, which undergoes shear deformation causing the large observed bending moments. It is clear that the liner is responsible for maintaining the integrity of the tunnel.

Additional Exercise

Repeat the previous analysis, but instead of applying a Composite Liner with a joint, apply a regular (single layer) liner. If you run the analysis, you will see the difference in the liner behaviour.

As you can see from the following figure, the Liner bending moment results are completely different from the Composite Liner (with joint) bending moments.



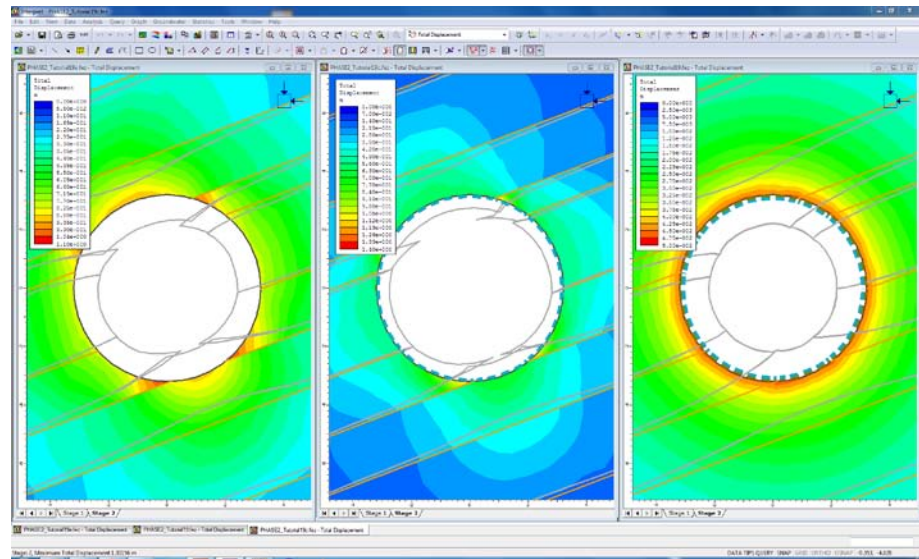
Liner bending moment (single layer liner, with no joint between liner and rock).

At the tunnel / joint intersections, the liner bending moments decrease to minimum values, rather than maximum values. This is because the liner is effectively discontinuous at these locations, and does not resist differential movement of opposite sides of each joint.

The reason that the Composite Liner (with joint) gives such different results from a single layer liner (with no joint), is due primarily to the way in which *Phase2* assigns node numbering at the intersections of joints. When a joint is present between the liner and the rock, this correctly models the physical interaction of the joints, tunnel boundary and liner.

Finally, the following figure illustrates the deformations for all three cases (unsupported, single liner, composite liner). NOTE: the scale factors used to display the deformed boundaries are as follows: unsupported (Scale Factor = 1), single liner (Scale Factor = 1), composite liner (Scale Factor = 20).

As you can see, the overall deformation for the single liner is not much different from the unsupported case. The differential movement at the joint ends is actually more pronounced for the single liner compared to the unsupported case. For the composite liner, the overall deformations are about 20 times less than the unsupported case (note scale factors), and the deformation pattern is relatively uniform and circular.



Deformed boundaries for (left to right) – unsupported, single liner, composite liner. Scale factor for deformations = 1, 1, 20, respectively.

This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.